Resolvent norm bound for adaptive Langevin dynamics with Nose-Hoover thermostatted friction, uniform in the friction and time-scale separation

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Editorial note. Journal numbering: the counted proposition prints as Proposition 3.12 in the publisher PDF (twelfth theorem-family item of Section 3: Assumption 3.1, Assumption 3.2, Theorem 3.3, Corollary 3.4, Proposition 3.5, Lemma 3.6, Lemma 3.7, Definition 3.8, Proposition 3.9, Remark 3.10, Proposition 3.11 precede it); the card seeds the section counter with 3 and the shared family counter cdrthm with 11, so the counted proposition prints as 3.12 exactly as in the journal.

Editorial note. Equation renumbering: the context's numbered equation eq:quadratic_U prints as (3.1) in the delivery baseline (journal (3.20) of Section 3); the counted statement and proof contain no numbered displays. All cross-references, including the proof's reference to the quadratic choice and the context's references to eq:Lham (journal (3.2), stubbed) and eq:LFD_Langevin (journal (3.3), stubbed), resolve through the two engine passes.

Editorial note. Citations: the delivered context cites LSS19, DMS15 and Herau06 (source lines 996 and 1008) and the counted proof cites LSS19 (Lemma 2.7) at source line 1026; the publisher bibliography is shipped whole and all keys resolve.

Editorial note. External references: the assumption ladder 2.1/2.2/2.3/2.5/2.6 and the input theorem thm:bounds_Linv (2.7) are stubbed as framework items; Assumption 3.1 is stubbed as the counted statement's hypothesis; Corollary cor:Lang (3.4) is stubbed for the proof's in-paper pointer to its computations; eq:Lham, eq:LFD_Langevin and the section anchor sec:abstract are stubbed for the context's references. None is counted by this card or by a sibling card.

Editorial note. Preamble composition: three ranges – the editorial numbering adapter first (family re-declaration on the cdrthm anchor counter, English names, numberwithin), then the retained preamble's provenance comments (lines 1-6) and its body (lines 8-99). Line 7, the active cedram documentclass, is excluded because the wrapper supplies the class; lines 100-160 (title, alttitle, authors, keywords, abstracts and front matter) are excluded as publisher class-metadata commands that carry no mathematical content. hyperref is declared through extra_packages; the wrapper's automatic amsmath, amssymb,

amsthm and graphicx load precedes the preamble, and the retained amssymb usepackage line is an inert duplicate. All notation macros used by the delivered spans are defined in the retained preamble itself; no mathematical macro, symbol, statement or proof step was added, changed or removed.

Editorial note. Occupancy and claim evidence: claimkey_index carries no ahl.129 key; the doi_public_occupancy gate sweep for doi 10.5802/ahl.129 ran PASS with occupied=0 in the 2026-10-15 morning gate over the full 30-candidate queue, and the census v3.12 row for this unit reads pairing own with occupied false.

Allowed standard prerequisites (editorial).

- The Section 2 abstract framework: the block decomposition and the assumptions ladder ass:structural_Pi (2.1), ass:A_inv (2.2), ass:macro_coercivity (2.3), ass:reversibility (2.5), ass:technical_ass (2.6) with the input theorem thm:bounds_Linv (2.7), all stubbed; the Langevin generators cLham (journal (3.2)) and cLFD (journal (3.3)) of Section 3.1; and the Nose-Hoover setup of Section 3.3 delivered as context. Section 2 framework, Section 3.1 generator definitions and Section 3.3 setup prose, outside or adjacent to the counted spans. Fixes the framework ladder the counted proof verifies and the dynamics the counted statement quantifies.

Source reference. Original reference 3.1: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 581–591. This stub points at Assumption 3.1 of the journal (source lines 581-591), a hypothesis of the framework rather than a result proved in the source; the stub carries the journal number for reference resolution and is not counted by this card or by a sibling card.

Source reference. Original reference 2.1: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 322–329. This stub points at the structural hypothesis of Section 2 (source lines 322-329), a hypothesis of the framework; not counted by this card or by a sibling card.

Source reference. Original reference 2.2: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 382–384. This stub points at the invertibility hypothesis of Section 2 (source lines 382-384), a hypothesis verified, not proved, by the counted proof; not counted by this card or by a sibling card.

Source reference. Original reference 2.3: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 392–397. This stub points at the macroscopic coercivity hypothesis of Section 2 (source lines 392-397), a hypothesis whose verification is part of the counted proof's work; the assumption statement itself is not counted by this card.

Source reference. Original reference 2.5: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 458–483. This stub points at the reversibility hypothesis of Section 2 (source lines 458-483), a hypothesis of the framework; not counted by this card or by a sibling card.

Source reference. Original reference 2.6: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 484–486. This stub

points at the technical boundedness hypothesis of Section 2 (source lines 484-486), a hypothesis whose verification is the core of the counted proof; the assumption statement itself is not counted by this card.

Source reference. Original reference 2.7: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 490–495. This stub points at the abstract framework theorem of Section 2 (source lines 490-495), which the counted proof applies at its opening and close; it is stated and proved in the source outside this card and is not counted here.

Source reference. Original reference 3.4: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 721–732. This stub resolves the counted proof's pointer 'see the proof of Corollary ...' (source line 1026); the corollary and its proof are not counted by this card or by a sibling card.

Source reference. Original equation 3.2: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 562–564. This stub resolves the delivered context's reference (source line 1004); a definition of the Section 3.1 setting, not counted by this card.

Source reference. Original equation 3.3: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 566–570. This stub resolves the delivered context's reference (source line 1004); a definition of the Section 3.1 setting, not counted by this card.

Source reference. Original reference 2: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 289–289. This stub resolves the delivered context's reference 'The framework introduced in Section 2' (source line 1008); a section anchor, not counted by this card.

$$U(p) = \frac{|p|^2}{2} = \frac{1}{2}\sum_{i=1}^{d} p_i^2. \tag{3.1}$$

After some suitable normalization (see [LSS19]), the invariant measure of the dynamics reads

$$\mu(dq\,dp\,d\xi) = Z^{-1}\exp\left(-\beta\left[\frac{|p|^2}{2} + V(q) + \frac{\xi^2}{2}\right]\right) dq\,dp\,d\xi,$$

and the generator is, for some $\varepsilon > 0$,

$$\mathcal{L} = \mathcal{L}_{\mathrm{ham}} + \gamma\mathcal{L}_{\mathrm{FD}} + \frac{1}{\varepsilon}\mathcal{L}_{\mathrm{NH}},$$

with $\mathcal{L}_{\mathrm{ham}}$ and $\mathcal{L}_{\mathrm{FD}}$ defined respectively in (3.2) and (3.3), and

$$\mathcal{L}_{\mathrm{NH}} = \left(|p|^2 - \frac{d}{\beta}\right)\partial_\xi - \xi\, p^T\nabla_p = \frac{1}{\beta^2}\left(\left(\partial_\xi - \partial_\xi^*\right)\nabla_p^*\nabla_p + \Delta_p^*\partial_\xi - \Delta_p\partial_\xi^*\right).$$

Therefore, $\mathcal{S} = \gamma\mathcal{L}_{\mathrm{FD}}$ as for standard Langevin dynamics, while $\mathcal{A} = \mathcal{L}_{\mathrm{ham}} + \varepsilon^{-1}\mathcal{L}_{\mathrm{NH}}$. The framework introduced in Section 2 allows to retrieve the scaling on the resolvent of the generator found in [LSS19] using the techniques from [DMS15, Herau06].

Proposition 3.12. *Suppose that Assumption 3.1 holds true. Then there exists $C \in \mathbb{R}_+$ such that*

$$\forall\, \gamma, \varepsilon > 0, \qquad \left\|\mathcal{L}^{-1}\right\| \leqslant C \max\left(\gamma\varepsilon^2, \gamma, \frac{1}{\gamma}, \frac{1}{\gamma\varepsilon^2}\right).$$

Proof. Assumptions 2.1, 2.2 and 2.5 hold true as for Langevin dynamics. A simple computation next shows that

$$\mathcal{A}_{+0}^* \mathcal{A}_{+0} = \left(\frac{2d}{\beta^2\varepsilon^2}\partial_\xi^*\partial_\xi + \frac{1}{\beta}\nabla_q^*\nabla_q\right)\Pi_0,$$

which, by tensorization of Poincaré inequalities, implies that Assumption 2.3 holds with

$$a^2 = \frac{1}{\beta}\min\left(\frac{2d}{\varepsilon^2}, K_\nu^2\right).$$

Let us next turn to Assumption 2.6. Computations similar to the ones performed for Langevin dynamics (see the proof of Corollary 3.4) show that $\Pi_1\mathcal{L}_{\mathrm{FD}}\Pi_1$ is bounded and $\mathcal{S}_{21}\mathcal{A}_{12}\Pi_0(B^*B)^{-1} = 0$ since $\mathcal{S}_{21} = \gamma\Pi_2\mathcal{L}_{\mathrm{FD}}\Pi_1 = 0$ for the choice (3.1). From the computations in the proof of [LSS19, Lemma 2.7], there exists $R \in \mathbb{R}_+$ such that

$$\left\|\Pi_+\mathcal{L}_{\mathrm{ham}}^2\Pi_0\left(\mathcal{A}_{+0}^*\mathcal{A}_{+0}\right)^{-1}\right\| \leqslant R\max\left(1, \frac{1}{\varepsilon}\right).$$

An application of Theorem 2.7 then allows to conclude. □

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